Multi-Node Routing Engine

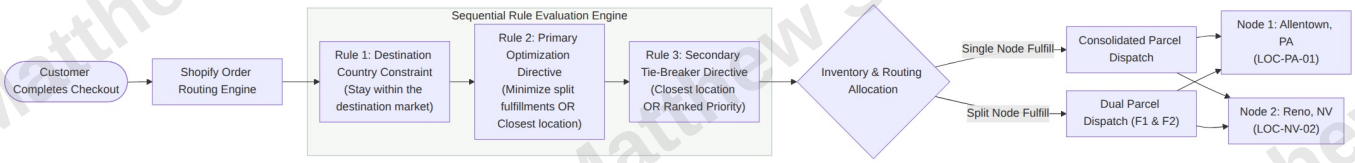
1. Executive Summary & Network Topology

Synapse Botanicals operates a distributed direct-to-consumer (DTC) fulfillment network across the contiguous United States. Orders are fulfilled dynamically using **Shopify's Native Order Routing Engine** (`Settings` → `Shipping and delivery` → `Order routing`).

This specification establishes:

1. **Network Node Architecture:** Distribution roles and SKU stocking profiles for East Coast and West Coast nodes.
2. **Algorithmic Evaluation Hierarchy:** The strict execution order of Shopify's routing rule stack.
3. **The Cross-Country Routing Trap:** Technical analysis of how rule priority alters parcel dispatch decisions.
4. **Safety Stock Buffer Thresholds:** Mathematical buffers preventing stockouts and unintentional split shipments.
5. **Operational Location Reassignment Protocols:** Procedures for manually overriding fulfillment nodes.

> **NOTE Routing Architecture & Empirical Simulation Reference:**
> The multi-node stocking profiles (`LOC-PA-01` Allentown, `LOC-NV-02` Reno) and routing algorithms specified in this document are benchmarked against simulated order cohorts in **CaseStudy - MultiLocationRoutingOptimization - Analysis (Domain 2.3)** using real UPS Ground freight rate cards.



2. Fulfillment Node Profiles & SKU Stocking Topology

To optimize delivery speed and reduce carrier shipping zones, inventory is partitioned across two strategic distribution centers:

2.1 Node Profiles

Node Identifier	Facility Name	Geographic Focus	Transit Zone Coverage	Primary Facility Role
`LOC-PA-01`	Allentown Distribution Center, PA	East Coast, Midwest, Mid-Atlantic	Zones 1–4 East	High-velocity consumable hub (Stick Packs & Starter Bundles)
`LOC-NV-02`	Reno Fulfillment Hub, NV	West Coast, Pacific NW, Southwest	Zones 1–4 West	Specialty tincture hub + hardware accessory stock + West consumable hub

2.2 SKU Stocking & Availability Matrix

> **NOTE Kitted inventory:** Bundles are stocked as pre-kitted units at both nodes in this routing topology. Reno-only loose tincture inventory does not force a pre-kitted bundle to Reno. Component-picked bundle scenarios must declare their alternate allocation/pick basis explicitly.

Not all SKUs are dual-stocked across both warehouses due to manufacturing batch sizes and storage requirements:

Product Title	SKU Identifier	Allentown, PA (`LOC-PA-01`)	Reno, NV (`LOC-NV-02`)	Stocking Strategy
Focus Stick Packs (Matcha Yuzu)	`SYN-FSTK-MY`	**Primary (High ATS)**	**Secondary (Buffer)**	Dual-Stocked (Universal velocity)
Focus Stick Packs (Blood Orange)	`SYN-FSTK-BO`	**Primary (High ATS)**	**Secondary (Buffer)**	Dual-Stocked (Universal velocity)
Focus Stick Packs (Wild Blueberry)	`SYN-FSTK-WB`	**Primary (High ATS)**	**Secondary (Buffer)**	Dual-Stocked (Universal velocity)
Focus Stick Packs (Crisp Green Apple)	`SYN-FSTK-GA`	**Primary (High ATS)**	**Secondary (Buffer)**	Dual-Stocked (Universal velocity)
Adaptogen Tincture (AM Focus)	`SYN-TNCT-AM`	— *(Not Stocked)*	**Exclusive Hub (100%)**	Single-Node Specialized Hub
Adaptogen Tincture (PM Restore)	`SYN-TNCT-PM`	— *(Not Stocked)*	**Exclusive Hub (100%)**	Single-Node Specialized Hub
The Daily Protocol Starter Bundle	`SYN-BNDL-MAO / SYN-BNDL-BAB`	**Primary (High ATS)**	**Secondary (Buffer)**	Dual-Stocked Kitted Unit
Precision Insulated Tumbler	`SYN-TMBL-MO / BW / SG`	**Primary Stock**	**Secondary Stock**	Dual-Stocked Hardware
Micro-Foam Electric Whisk	`SYN-ACC-WHSK`	— *(Not Stocked)*	**Exclusive Hub (100%)**	Single-Node Accessory Stock

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3. Algorithmic Routing Rule Hierarchy

Shopify evaluates order routing rules **sequentially from top to bottom**. Each rule acts as a filter or a sort function on the list of eligible fulfillment locations.

3.1 Production Rule Stack Configuration

1. Minimize split fulfillments.
2. Stay within the destination market.
3. Ship from the closest location.

Use ranked locations or location-metafield rules only as explicitly configured additional preferences.

3.2 Detailed Execution Mechanics

Rule 1: Market preference

The native market rule prioritizes locations in the destination market; it is not a hard sovereign-border filter. Any hard domestic-only constraint requires a separate documented configuration.

> **NOTE Tags and custom routing:** `flow-route:*` tags are metadata. They change allocation only when a named custom rule, middleware consumer or operator reads them and performs a supported action. Order-created tagging can occur after initial routing, so do not assume it precedes assignment.

Rule 2: Minimize split fulfillments (Primary Optimization)

- **Mechanics:** Evaluates all possible combinations of candidate locations that can fulfill the entire line item set:
  - **Option A (Single Location Candidate):** If any single warehouse contains $100%$ of the ordered items in Available-to-Sell (ATS) stock, Shopify **discards all multi-location combinations**.
  - **Option B (Multi-Location Fallback):** If no single warehouse has all items, the engine calculates the minimum number of locations required (e.g., 2 locations instead of 3).

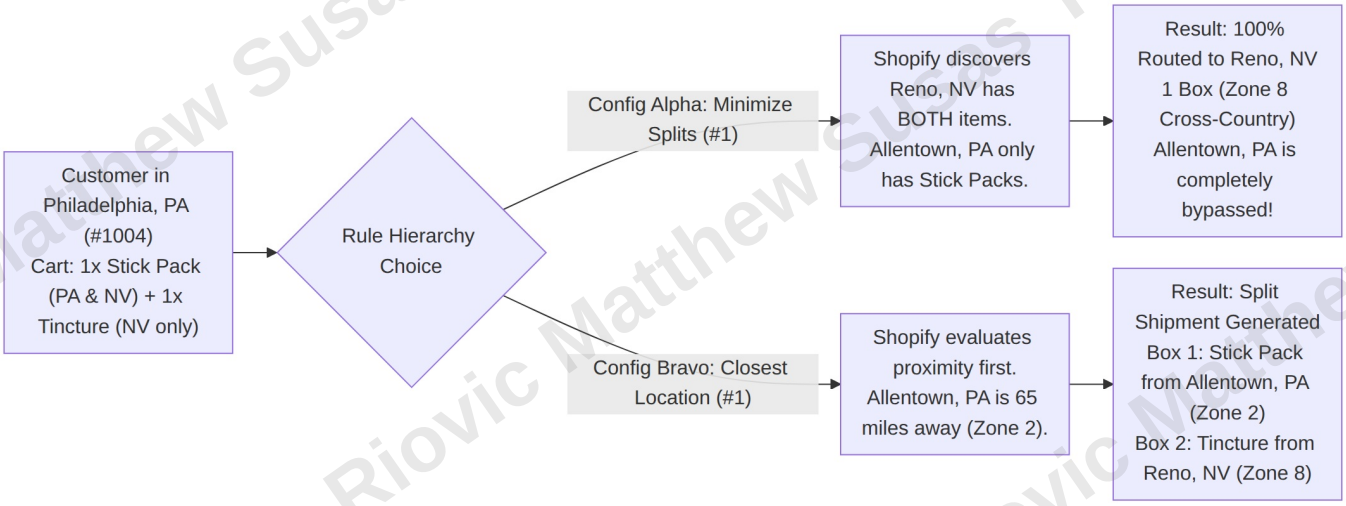
Rule 3: Ship from closest location (Distance Sort & Tie-Breaker)

- **Mechanics:** Calculates straight-line geodesic distance between the customer's delivery ZIP code and the geocoded address of eligible candidate locations.
- **Tie-Breaker:** When two or more single-location candidates can fulfill the entire order, the engine selects the facility closest to the recipient.

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4. The "Hidden Gap" Exposed: Split-Averse vs. Proximity Routing Trap

The most common operational failure in multi-warehouse e-commerce is the **Cross-Country Routing Trap**. When operators do not understand how Rule 2 interacts with Rule 3, they mistakenly believe the warehouse integration is defective.



4.1 Comparative Scenario Analysis

Test Order Parameters:

- **Customer Location:** Philadelphia, PA (`19102`)
- **Basket Items:**
  - 1x `Focus Stick Packs (30-Pack)` – Dual-stocked at `LOC-PA-01` and `LOC-NV-02`
  - 1x `Adaptogen Tincture (60ml)` – Stocked **exclusively** at `LOC-NV-02`

System Metric	Configuration Alpha (Split-Averse: Production)	Configuration Bravo (Proximity-Averse: Split Allowed)
Rule Order	1. Minimize split fulfillments 2. Closest location	1. Ship from closest location 2. Minimize split fulfillments
Assigned Warehouse(s)	`LOC-NV-02` **(Reno, NV) exclusively**	`LOC-PA-01` **(PA) &** `LOC-NV-02` **(NV) [SPLIT]**
Packages Dispatched	**1 Package** (Both items consolidated)	**2 Packages** (Dual split packages)
Carrier Zone(s)	Zone 8 (Cross-country: Reno → Philly)	Package 1: Zone 2 (Allentown → Philly) Package 2: Zone 8 (Reno → Philly)
Direct Fulfillment Cost	**$10.85** in the isolated 0.94 lb routing fixture	**$18.00** ($7.65 PA + $10.35 NV); $7.15 higher

| **Transit Time (SLA)** | 4–5 Business Days (Ground) | Stick Pack: 1–2 Days | Tincture: 4–5 Days | | **Customer Experience** | 1 Box, 1 Tracking Number | 2 Boxes arriving on different days | | **Operator Perception** | "*Why did PA get skipped? Is PA broken?*" | "*Why did this split? The split costs more in this fixture!*" |

> **IMPORTANT THE PRODUCTION DIRECTIVE**
> Synapse Botanicals maintains **Configuration Alpha** as the production baseline. Ingestible botanical drink mixes and tinctures are routinely consumed together as a morning ritual. Consolidating into a single package saves **$7.15 in modeled direct fulfillment cost for the routing fixture pair** and eliminates customer confusion caused by staggered delivery dates.

> **NOTE Empirical Validation Reference:**
> For an end-to-end audit trail of how this exact split-routing scenario was processed, intercepted, and reconciled with live customer exceptions, review **Audit - OrderStateSimulation - RunSheet (Domain 1.3)**.

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5. Inventory Buffers & Safety Stock Governance

To prevent stockouts from high-velocity sales spikes and avoid race conditions between Shopify and 3PL WMS synchronization cycles, safety stock buffers must be established.

5.1 Buffer Mathematics & Available-to-Sell (ATS) Logic

Model Shopify inventory states separately from any external WMS buffer:

$Available = On Hand - Committed - Unavailable$

Safety stock is part of Unavailable when represented there. If a separate WMS-only buffer is used, document its mapping; do not subtract it again from Shopify Available.

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│ INVENTORY SAFETY BUFFER SPECIFICATION                                    │
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│ Product Category         │ Velocity Tier│ PA Buffer    │ NV Buffer    │ Rationale    │
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│ Focus Stick Packs (All SKUs) │ High (Tier 1)│ 15 Units     │ 10 Units     │ Rapid spikes │
│ Adaptogen Tinctures      │ Med  (Tier 2)│ —            │ 10 Units     │ Single-hub   │
│ Starter Protocol Bundles │ Med  (Tier 2)│ 10 Units     │ 5 Units      │ High AOV kit │
│ Tumblers & Whisks        │ Low  (Tier 3)│ 5 Units      │ 5 Units      │ Stable drain │
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5.2 Dynamic Out-of-Stock Fallback Cascade

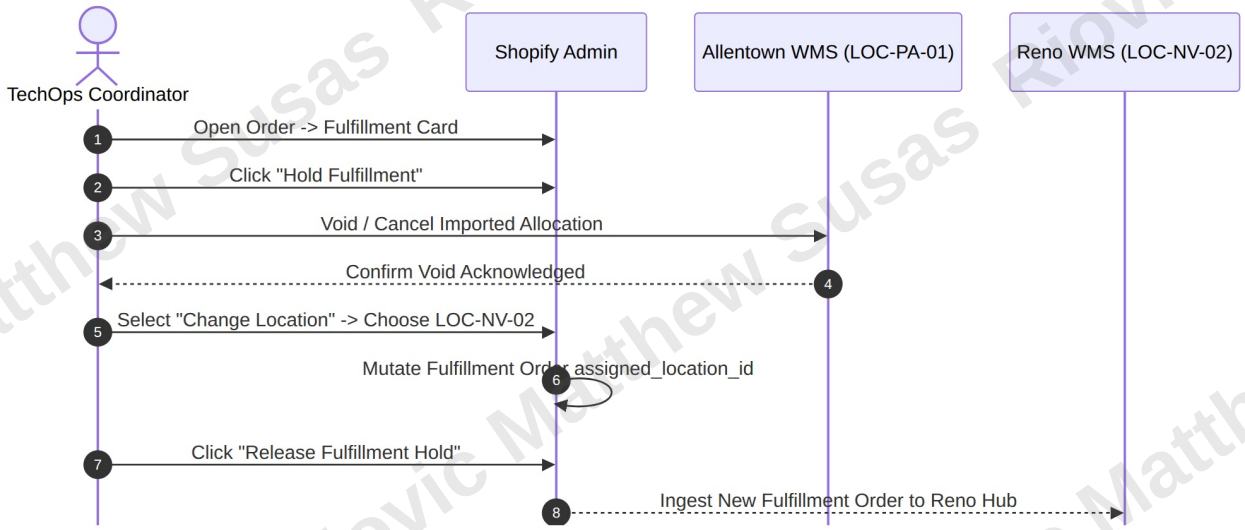
When a distribution node's ATS reaches \le 0 (due to on-hand inventory hitting the safety buffer):

1. **Automated De-allocation:** The native routing engine automatically disqualifies that location from candidate pools for that SKU.
2. **Future assignment:** Eligible new orders can use the secondary node. Existing fulfillment orders require a supported move/reassignment and source-WMS cancellation acknowledgment; stock changes alone do not guarantee automatic transfer.
3. **Low-Stock Notification Trigger:** If total network ATS falls below 25 units, Shopify Flow generates an automated Slack notification to `#supply-chain-alerts`.

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6. Operational Location Reassignment Protocol (Manual Overrides)

When physical warehouse constraints (e.g., weather delays, carrier trailer shortages, physical inventory audits) require an operational override, TechOps coordinators must execute a manual location reassignment.



6.1 Step-by-Step Location Override Procedure

> **NOTE System Governance & Safety Dependency:**
> Location reassignments must strictly follow the native hold protocol in **SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)**, and are strictly prohibited once an order crosses the Point of No Return defined in **Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)**.

1. **Place Fulfillment Hold:** Never change locations on an active, unheld fulfillment order. Apply hold code: `Inventory issue`.
2. **Void Original 3PL Record:** Access the source warehouse portal (e.g., ShipBob Allentown). Search Order ID → Click **Cancel / Void Order**. Confirm the shipment is not in a wave pick.
3. **Execute Location Mutation in Shopify Admin:**
   - Navigate to the target fulfillment card.
   - Click the **three horizontal dots (...)** → Select **Change location**.
   - In the location dropdown, select the target facility (e.g., `Reno Fulfillment Hub`).
   - Verify line items and quantities transfer completely.
   - Click **Save**.
4. **Release Hold:** Release the Shopify fulfillment hold. Confirm the order transmits to the destination 3PL queue through the configured fulfillment integration; a hold release is not a scheduling event.
5. **Tag Order:** Append tracking tag: `OPS: Rerouted [Old_Loc] to [New_Loc]`.

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7. Change Management & Verification Checklist

Modifications to the Shopify Order Routing rule stack can affect subsequent eligible order assignments. Any proposed changes to the rule order must undergo the following release gate:

- [ ] **Off-Peak Execution:** Routing rule adjustments must be deployed Tuesday–Thursday between 10:00 PM and 11:30 PM EST.
- [ ] **Pre-Flight Test Suite:** Run 4 synthetic draft orders representing:
    1. East Coast Single-Node basket (PA only)
    2. West Coast Single-Node basket (NV only)
    3. East Coast Mixed basket (PA & NV SKUs)
    4. Out-of-Stock Fallback basket (PA inventory depleted)
- [ ] **3PL Ingestion Validation:** Confirm test order fulfillment orders are ingested into the correct warehouse WMS tenant within 180 seconds.
- [ ] **Post-Deploy Audit:** Review the first 25 live production orders following deployment to ensure fulfillment card locations match the expected model.

Technical references — checked 30 October 2026

- Shopify routing rules
- Shopify inventory states

Riovic Matthew Susas